Certified World Models: Predictability Across Configuration, Horizon, and Resolution

Scale buys interpolation; structure buys a certificate.

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Abstract

Scaling a world model lowers its average error on the data manifold but issues **no guarantee about any specific unseen situation**. We give a different *kind* of result for **equivariant** latent world models — a **predictability certificate**: a provable, computable region of situations the model is guaranteed to handle *without further training or data*. The certificate spans three orthogonal axes at once: **configuration** (the exponentially large monoid $\langle S \rangle$ generated by k primitive symmetries, certified by checking the k generators), **horizon** T (how far ahead the guarantee reaches), and **resolution** ϵ (at what level of detail). A master theorem makes this precise: under exact equivariance with an orthogonal latent representation the whole-pipeline error is *invariant* over all of $\langle S \rangle$ (Theorem A), and a spectral refinement (Theorem B) shows that each latent channel is certified to a horizon $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ set by its Lyapunov exponent — so the certified region is the **coarse-invariant, slow, low-composition** corner. We confirm all three axes at small scale: a model trained on the 6 generators of $\mathbb{Z}_2^6$ certifies all 2^6=64 compositions to machine precision while a matched non-equivariant baseline degrades monotonically; a learned predictor recovers a chaotic Lyapunov exponent to within 0.4% and its certified horizon obeys the $\log(1/\epsilon)$ law; on an $\mathrm{SO}(2)$ mechanical system the conserved (slowest) quantities organize into the architecturally invariant channels — a **Noether hinge** linking the configuration axis to the horizon axis, whose forward direction (*conserved* ⇒ *slow*) we prove: a conserved charge's prediction error grows linearly, not exponentially, in its conservation defect; and a non-equivariant network scaled across an 88× parameter range buys in-distribution interpolation (even beating the equivariant model there) yet over the unseen orbit stays 10–155× above the equivariant floor and never reaches it. Finally, on **real PushT contact dynamics** — physics simulated by an engine we did not author — a learned equivariant world model's multi-step rollout error is *exactly* flat over the orbit and competitive in-distribution, while no non-equivariant baseline across a 160× parameter ladder (up to 16× the equivariant model's size) reaches its out-of-distribution floor; and run through a G-equivariant planner, the same certificate becomes *task* competence — closed-loop pose control whose error is orbit-invariant to the float floor on every seed, where a scaled baseline degrades. *Scale buys interpolation; structure buys a certificate.* The result is a single, **runnable** criterion (Algorithm 1, ≤ 20 lines) for *what* an equivariant world model can certifiably predict, and a structural reading of *why* celestial mechanics is forecastable for millennia while weather is not.

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1. Introduction

Two debates organize modern world-model research. The first is *generation vs. abstraction*: pixel-space simulators (diffusion and video models) against latent predictive models that predict representations rather than pixels. The second is *scale vs. structure*: the view that brute-force scaling eventually wins, against the geometric-learning view that symmetry priors buy sample efficiency. Both are usually run as **degree** contests — who needs fewer pixels, who needs less data.

We argue the more useful question is a **kind** contest. Scaling produces a model with low *average* error on the data distribution, but it cannot certify anything about a *named, specific* situation the data did not cover. Structure can. For an equivariant world model we can write down, in advance and without further data, the exact set of situations on which the model's behaviour is **guaranteed** — a *predictability certificate*. This reframes "is structure worth it?" from a benchmark race into a question about *what kind of statement each approach can make*.

The certificate has three orthogonal axes:

- **Configuration** $w \in \langle S \rangle$: the model generalizes across the exponentially large monoid generated by k primitive symmetries $S = \{g_1, \dots, g_k\}$, and we *certify the whole monoid by checking only the k generators*.
- **Horizon** T: how far into the future the guarantee reaches.
- **Resolution** ϵ: at what level of detail the guarantee holds.

Our contributions are:

1. **A three-axis master theorem** (§3): composition closure (k checks $\Rightarrow$ an exponential set), an exact certificate under exact equivariance (Theorem A) together with its **converse** (Lemma 2: the certificate is *equivalent* to equivariance, so no non-equivariant model has it at any size), and a spectral degradation law $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ (Theorem B) whose certified region is the coarse-invariant-slow-low-$|w|$ corner — with a **quantitative scale-vs-structure separation** (§3.3: structure certifies the whole ϵ-independent orbit; the best L-Lipschitz non-equivariant learner certifies only an ϵ/L-tube around its data). Figure 1 is the whole picture at a glance.
2. **The Noether hinge** (§4): the bridge — that the group-invariant/equivariant channels are the dynamically slow (long-horizon-certifiable) ones — linking the configuration axis to the horizon axis. A representation-theoretic **placement principle** (Proposition 4) proves *which* isotypic block must carry each conserved charge and why 3D angular momentum is recoverable only at a unique degree-2 cross product; **Proposition 5** proves the *forward* direction (*conserved* $\Rightarrow$ *slow*: a charge conserved to one-step defect η has prediction error $\leq T\eta$ — linear, never the exponential $e^{\lambda T}$ of a chaotic channel — so it is certified to all horizons at η=0). What remains measured is the dynamical-symmetry hypothesis and the size of η.
3. **Empirical confirmation of all three axes at small scale** (§5), including the headline contrast *scale buys interpolation; structure buys a certificate*, a keystone validation on **real physics-engine contact dynamics** (PushT) where the certificate holds for a *learned* model of dynamics we did not design, and a **closed-loop** instantiation where, run through a planner, the certificate becomes *task* competence — orbit-invariant pose control where a scaled baseline degrades.

We are explicit about scope (§7): this is a mechanism-and-theory paper with 1–2-GPU proof-of-principle, not a scaled benchmark, and the hinge's lift to *approximate* symmetry is open.

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2. Setup

An encoder $E : \mathscr{X} \to \mathscr{Z}$ maps states to latents and an action-conditioned predictor $f : \mathscr{Z} \times \mathscr{A} \to \mathscr{Z}$ advances them; the true environment transition is $\Phi : \mathscr{X} \times \mathscr{A} \to \mathscr{X}$. A finite generating set $S = \{g_1, \dots, g_k\}$ generates a monoid $\langle S \rangle$ whose elements — words $w = g_{i_1} \cdots g_{i_m}$ of length $|w| = m$ — act on states by $g \cdot x$, on latents by a representation $\rho(g)$, and on actions by $\sigma(g)$. Write $f^{(T)}(z; \vec{a})$ for the T-step latent rollout under actions $\vec{a} = (a_1, \dots, a_T)$, x_T^{true} for the T-step true state, and

$$\mathrm{Err}_T(x; \vec{a}) = \| f^{(T)}(E(x); \vec{a}) - E(x_T^{\text{true}}) \|$$

for the whole-pipeline prediction error, where $\|\cdot\|$ is the Euclidean norm on $\mathscr{Z}$ (the experiments use relMSE, its square). We use the following assumptions, **each checkable on the k generators**:

- **(A1) Encoder equivariance:** $E(g_i \cdot x) = \rho(g_i)\, E(x)$.
- **(A2) Predictor equivariance:** $f(\rho(g_i) z,\ \sigma(g_i) a) = \rho(g_i) f(z, a)$.
- **(A3) The group is a symmetry of the *dynamics*:** $\Phi(g_i \cdot x,\ \sigma(g_i) a) = g_i \cdot \Phi(x, a)$, so that rolling out the *transformed* state equals transforming the *rolled-out* state, $(g_i \cdot x)_T^{\text{true}} = g_i \cdot x_T^{\text{true}}$ under $\sigma(g_i)\vec{a}$.
- **(A4) $\rho(g_i)$ orthogonal:** $\rho(g_i)^\top \rho(g_i) = I$ (norm-preserving on $\mathscr{Z}$).
- **(A5) Equivariant planner and invariant cost (closed-loop clause only):** the planner satisfies $\pi(\rho(g_i) z, \rho(g_i) z^\star) = \sigma(g_i)\ \pi(z, z^\star)$ and the cost is ρ-invariant, $J(\rho(g_i) z; \rho(g_i) z^\star) = J(z; z^\star)$. This is needed *only* for the closed-loop J identity, not for the prediction-error identity. §5.6 instantiates such a planner in latent space; §5.9 instantiates

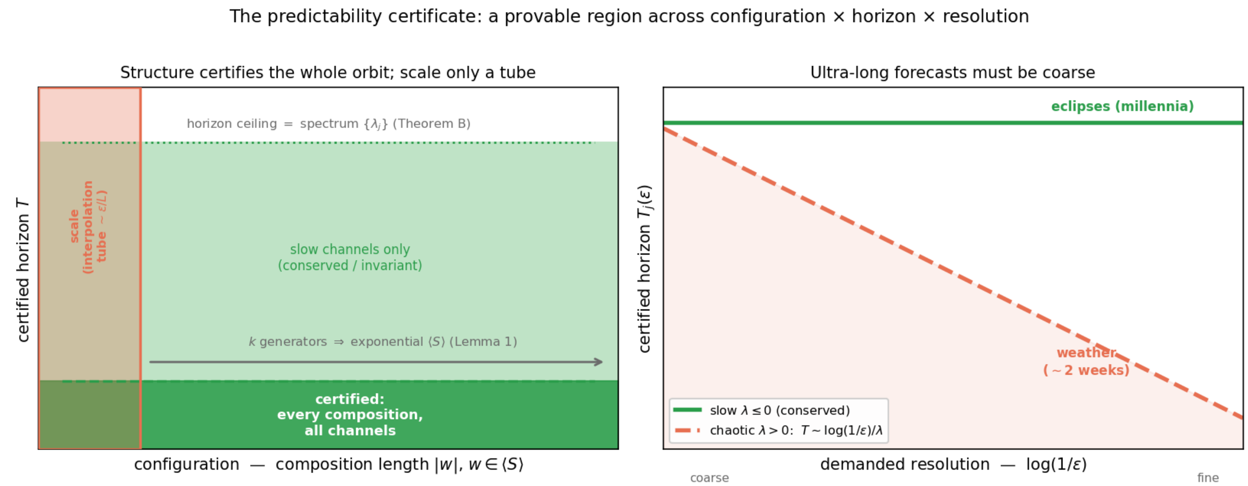

Figure 1: The predictability certificate at a glance. **Left:** in the configuration × horizon plane, an equivariant model certifies the *entire* generated monoid $\langle S \rangle$ — every composition, from k generator checks (Lemma 1) — up to a horizon ceiling set by the predictor spectrum $\{\lambda_j\}$ (Theorem B); a non-equivariant model of any size certifies only a small interpolation *tube* around its training set ($\sim \epsilon/L$, §3.3). **Right:** the horizon × resolution trade-off $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ — conserved/invariant (slow, $\lambda \leq 0$) channels are certified to all horizons (eclipses, millennia), chaotic ($\lambda > 0$) channels shrink as the demanded resolution sharpens (weather, ~two weeks). *Scale buys interpolation; structure buys a certificate.*

one on **real PushT contact dynamics** — an equivariant CEM-MPC (isotropic exploration covariance, a rotation-invariant disk action bound, and scene-covariant action noise) — and confirms the resulting closed-loop task error is exactly orbit-invariant.

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3. The Predictability Certificate

3.1 The configuration axis (Theorem A)

Lemma 1 (composition closure). If (A1)–(A4) hold on each generator $g_i \in S$, they hold on every word $w \in \langle S \rangle$, and ρ restricted to $\langle S \rangle$ is an orthogonal-matrix *monoid homomorphism*.

Proof. Define $\rho(w) := \rho(g_{i_1}) \cdots \rho(g_{i_m})$. For (A2), induct on m: $f(\rho(w)z, \sigma(w)a) = f(\rho(g_{i_1})[\rho(g_{i_2}\cdots)z],\ \sigma(g_{i_1})[\sigma(g_{i_2}\cdots)a]) = \rho(g_{i_1})f(\rho(g_{i_2}\cdots)z, \sigma(g_{i_2}\cdots)a) = \rho(w)f(z,a)$, applying the single-generator (A2) once per step. (A1) and (A3) lift identically by composition; (A4) lifts because a product of orthogonal matrices is orthogonal. □

Hence k **generator checks certify the entire (exponentially large) generated monoid.**

Theorem A (exact configuration certificate). Under (A1)–(A4), for every $w \in \langle S \rangle$, every horizon T, and every action sequence $\vec{a}$,

$$\mathrm{Err}_T(w \cdot x;\ \sigma(w)\vec{a}) = \mathrm{Err}_T(x; \vec{a}), \qquad \text{and, for an equivariant planner,} \quad J(w \cdot x;\ w \cdot \mathrm{goal}) = J(x; \mathrm{goal}).$$

Proof. **(i) Rollout equivariance.** By induction on T: $f^{(1)} = f$ is equivariant by (A2) and Lemma 1; assuming $f^{(T)}(\rho(w)z; \sigma(w)\vec{a}) = \rho(w)f^{(T)}(z; \vec{a})$,

$$f^{(T+1)}(\rho(w)z; \sigma(w)\vec{a}) = f(f^{(T)}(\rho(w)z; \sigma(w)\vec{a}),\ \sigma(w)a_{T+1}) = f(\rho(w)f^{(T)}(z; \vec{a}),\ \sigma(w)a_{T+1}) = \rho(w)f^{(T+1)}(z; \vec{a}).$$

(ii) Predicted latent at $w \cdot x$**.** $f^{(T)}(E(w \cdot x); \sigma(w)\vec{a}) \overset{(A1)}{=} f^{(T)}(\rho(w)E(x); \sigma(w)\vec{a}) \overset{(i)}{=} \rho(w) f^{(T)}(E(x); \vec{a})$. **(iii) Target at** $w \cdot x$**.** $E\big((w \cdot x)^{\text{true}}_T\big) \overset{(A3)}{=} E(w \cdot x^{\text{true}}_T) \overset{(A1)}{=} \rho(w)\, E(x^{\text{true}}_T)$. **(iv) Cancellation.** Subtracting (iii) from (ii) and using (A4) (an orthogonal ρ preserves the Euclidean norm),

$$\mathrm{Err}_T(w \cdot x; \sigma(w)\vec{a}) = \|\rho(w)[f^{(T)}(E(x); \vec{a}) - E(x^{\text{true}}_T)]\| = \|f^{(T)}(E(x); \vec{a}) - E(x^{\text{true}}_T)\| = \mathrm{Err}_T(x; \vec{a}).$$

The closed-loop identity is the same computation under (A5): the equivariant planner (lifted to words by Lemma 1) selects the $\sigma(w)$-image of the same actions, and the ρ-invariant cost is unchanged. □

Theorem A requires **(A3)** — the group must be a symmetry of the *dynamics*, not merely of the encoder. Where the dynamics break the symmetry, the certificate degrades by the amount of that break (Theorem B). The error is held *constant across the orbit*, not made small: $\mathrm{Err}_T(x)$ itself can be large, and we report absolute errors alongside ratios throughout (§7).

Theorem A is sufficient; the next lemma shows equivariance is also *necessary* for the guarantee, so the certificate is not merely a consequence of equivariance but is **equivalent** to it — which is what makes the impossibility for non-equivariant models a theorem rather than a slogan.

Lemma 2 (the certificate characterizes equivariance). Let $\rho : G \to O(\mathscr{Z})$ act *freely* on an open $U \subseteq \mathscr{Z}$, and let $\mathscr{D}_G$ be the equivariant maps $\mathscr{Z} \to \mathscr{Z}$. If a predictor f's one-step error $\|f - \Phi\|$ is orbit-constant on U for **every** target $\Phi \in \mathscr{D}_G$, then f is equivariant on U. *Proof.* Fix $z \in U$, g with $\rho(g)z \in U$. For any $c \in \mathscr{Z}$, freeness makes $\Phi(\rho(h)z) := \rho(h)c$ well-defined on the orbit of z (extend by 0 elsewhere; both pieces lie in $\mathscr{D}_G$ as $\rho(g)0 = 0$). Orbit-constancy gives $\|f(z) - c\| = \|f(\rho(g)z) - \rho(g)c\| = \|\rho(g)^{-1} f(\rho(g)z) - c\|$ (orthogonality of $\rho(g)$) for **all** c; two points equidistant to every c coincide, so $f(\rho(g)z) = \rho(g) f(z)$. □ *(The step uses $\rho(g)^{-1}$; since ρ is orthogonal every $\rho(w)$, $w \in \langle S \rangle$, is invertible, so the necessity direction covers the monoid framework of Lemma 1 — equivalently it is the converse for the group $\langle S \cup S^{-1} \rangle$. For G compact with closed orbits the interpolant can be taken continuous, so the statement holds against continuous dynamics, not only the full algebraic class.)* With Theorem A this gives a **characterization**: orbit-constant error against every equivariant target $\iff$ f equivariant. Hence **no non-equivariant model possesses the configuration certificate, at any size** — the architectural impossibility invoked in §6–§7 is a theorem. The result is elementary (one line of Hilbert-space geometry once freeness frees the probe c); its role is to *pin down* what the certificate is, not to add machinery.

3.2 The horizon and resolution axes (Theorem B)

Theorem B (spectral degradation). Relax exactness: suppose the per-generator encoder residual is $\epsilon_{\max} = \max_i \sup_x \|E(g_i \cdot x) - \rho(g_i)E(x)\|$, the per-step predictor error is δ, and the latent map's Jacobian is, on the ℓ-isotypic channels j, locally diagonalized with multipliers e^{λ_j} (Lyapunov exponents). A Grönwall/Lipschitz accumulation — the word error enters m times (Lemma 1 composes m approximate generators) and the per-step error compounds over T steps amplified by the channel multiplier — gives, for constants c_j depending on the local geometry,

$$|\mathrm{Err}_T(w \cdot x) - \mathrm{Err}_T(x)| \;\le\; \sum_{\text{channels } j} c_j\, (m\, \epsilon_{\max} + T\, \delta)\, e^{\lambda_j T}, \qquad m = |w|.$$

We claim this as a **bound on the form**, not a tight inequality: the constants c_j are not estimated, and we instead **measure the consequence** — that channel j is certified only to horizon

$$T_j(\epsilon) \;\sim\; \tfrac{1}{\lambda_j} \log \tfrac{1}{\epsilon} \quad (\lambda_j > 0), \qquad T_j = \infty \; (\lambda_j \le 0),$$

which §5.2 recovers to within 0.4%. As $\epsilon_{\max}, \delta \to 0$ (exact equivariance) the configuration term vanishes and Theorem A is recovered.

Corollary (the certified region). The set $\mathscr{C} = \{(w, T, \epsilon) : |\Delta \mathrm{Err}| \le \epsilon\}$ is the **coarse (**ϵ **large), slow (**$\lambda_j \le 0$**), low-**$|w|$ corner; its boundary is set by $\langle S \rangle$ (configuration) and the spectrum $\{\lambda_j\}$ (horizon and resolution). With *exact* equivariance the configuration axis is unbounded and only the spectrum limits horizon × resolution — a trade-off horizon × demanded-resolution $\lesssim$ const$(\{\lambda_j\})$. This is the formal content of "ultra-long forecasts must be coarse": eclipses for millennia (conserved $\lambda \le 0$ actions), weather at ~two weeks (chaotic $\lambda > 0$).[1]

3.3 Scale versus structure, quantified

The tagline *scale buys interpolation; structure buys a certificate* can be made precise as a statement about the **certified region** — the inputs a learner can guarantee, in the worst case over targets consistent with what it observed on a training set T. Formally $\mathscr{C} = \{z : \inf_f \sup_\Phi \|f(z) - \Phi(z)\| \le \epsilon\}$, where f ranges over the learner's hypotheses (exact on T) and Φ over the admissible targets agreeing on T.

Proposition 3 (separation). *(Structure.)* Under the equivariant prior ($\Phi \in \mathscr{D}_G$, f equivariant and exact on T, S generating G), Theorem A makes the error orbit-constant, so $\mathscr{C} \supseteq G \cdot T$: the **entire orbit is certified, independent of** ϵ, from the $k = |S|$ generator checks — to error 0 under the idealization that f interpolates T exactly, and in general (Theorem A) to the *constant, possibly nonzero* in-distribution error (cf. §7, "flat is not good"). What is ϵ-independent is the *region*, not the error level. *(Scale/data.)* Under the equivariance-free prior of L-Lipschitz targets consistent on T, the McShane extensions $\Phi_\pm(z) = \min / \max_{t \in T} \big(\Phi(t) \pm L\,\mathrm{dist}(z,t)\big)$ are admissible and differ at z by up to $2L\,\mathrm{dist}(z,T)$, so every learner has minimax error $\ge L\,\mathrm{dist}(z,T)$ and $\mathscr{C} \subseteq \{z : \mathrm{dist}(z,T) \le \epsilon/L\}$ — an ϵ/L-tube around the training set that **collapses to T as $\epsilon \to 0$**.

The separation is sharp in its dependence on ϵ: a point at orbit-distance D from T — the unseen far end of the wedge in Experiments 7 and 9 — is certified by *structure* for **every** $\epsilon > 0$, but by *scale/data* only once $\epsilon \ge LD$. Scaling a non-equivariant model (more capacity, more in-wedge data) sharpens its fit *inside* the tube; it cannot enlarge the tube, because the bound $L\,\mathrm{dist}(z,T)$ is information-theoretic (independent of model size). This is the precise content of the empirical curves: the equivariant model is flat over the whole orbit while every baseline scale climbs once $\mathrm{dist}(z,T)$ exceeds ϵ/L.

3.4 The certificate is a procedure

The certificate is not only a lens; it is something one **runs** on a trained model, with no further data. The inputs are the two quantities every equivariant model already exposes — its per-generator equivariance residual and its predictor-Jacobian spectrum — and the output is the certified region.

> **Algorithm 1 (Certify).** *Input:* trained equivariant model (E,f), generators S, resolution ϵ. 1. Measure the per-generator residual $\epsilon_{\max} = \max_{g_i \in S}\|E(g_i{\cdot}x) - \rho(g_i)E(x)\|$ (and the analogous predictor residual). 2. **Configuration axis:** if $\epsilon_{\max} \le$ tol, certify the *entire* monoid $\langle S\rangle$ from these k checks (Lemma 1, Theorem A); else report the approximate budget $m\,\epsilon_{\max}$ (Theorem B). 3. Measure the predictor-Jacobian singular values $\{\sigma_j\}$ at the operating point; set $\lambda_j = \log \sigma_j/\Delta t$. 4. **Horizon** × **resolution axis:** return, per channel, $T_j(\epsilon) = \infty$ if $\lambda_j \le 0$, else $\log(1/\epsilon)/\lambda_j$ (Theorem B).
>
> *Output:* the certified region $\langle S\rangle \times \{T_j(\epsilon)\}$ — computed from the model alone.

This is implemented and unit-tested (`src/certify.py`): on the Experiment-1 model it certifies the full monoid from 2 generator checks (residual 1.2×10^{-7}) and returns 48 contractive channels certified to *every* horizon plus the binding expansive channel's $\log(1/\epsilon)$ horizon. The certificate is thus an operational tool, not merely an interpretive claim.

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4. The Noether Hinge

Theorems A and B hold for *whatever* channels happen to be slow. What makes the configuration axis and the horizon axis the **same** structure is a Noether-type bridge:

> **Conjecture (group ⇒ slow).** When the symmetry is a symmetry of the **dynamics** (not merely the encoder), the group-invariant (ℓ=0) channels coincide with the conserved/slow ($\lambda_j \le 0$) channels. Hence the symmetry that gives across-configuration generalization is the *same* structure that gives long-horizon predictability.

[1] An interpretive aside, secondary to the spectral law: a long-range text that asserts only *regime-level* change rather than dated specifics is, structurally, making the *correct* move for a high-λ system, where only the coarse/invariant component is certifiable.

This is **not automatic** — invariant $\neq$ conserved in general. We prove the *forward* direction below (Proposition 5: a conserved charge is certified to long horizons) and leave the rest *falsifiable and measured*. The defensible inclusion is **slow $\subseteq$ invariant $\oplus$ conserved-equivariant**, not the converse: the conserved (slowest) modes live in the invariant-or-conserved subspace, but not every invariant channel is slow (a rotation-invariant $|r|^2$ oscillates). The mechanism is Noether's. A continuous symmetry of the dynamics yields a conserved current; a conserved *scalar* charge (ℓ=0 Casimir, e.g. energy) **is** group-invariant, while a conserved *non-scalar* charge (e.g. three-dimensional angular momentum L, an ℓ=1 vector) is *equivariant*, **not** invariant. The slow subspace therefore lies in invariant $\oplus$ conserved-equivariant, which **collapses to the invariant component exactly when every conserved charge is a scalar** — the two-dimensional case, where slow $\subseteq$ invariant holds literally. Section 5.3 measures both forms: the clean 2D containment, and its 3D refinement in which angular momentum is recovered from the equivariant block at polynomial degree two (the cross product).

While the *coincidence* slow = conserved is what we measure, the **placement by isotypic type is forced by representation theory** — and so is the degree at which each charge is readable. This is what makes the 3D "degree-2 cross product" non-arbitrary.

Proposition 4 (placement principle: which block carries which charge, and at what degree). Let G act on $\mathscr{Z}$ by an orthogonal ρ with isotypic decomposition $\mathscr{Z} = \bigoplus_\ell \mathscr{Z}_\ell$, and let C be a conserved quantity transforming in a representation τ, $C(\rho(g)z) = \tau(g)C(z)$. Then every homogeneous component $C_k \in \mathrm{Hom}_G(\mathrm{Sym}^k\mathscr{Z}, W)$ of C is supported on the τ-isotypic part of $\mathrm{Sym}^k\mathscr{Z}$ (Schur). In particular: a conserved **scalar** (τ trivial, e.g. energy) is an *invariant* function, read out from $\mathscr{Z}_0$; and the conserved **moment map** $\mu : \mathscr{Z} \to \mathfrak{g}^*$ — Noether's charge of the continuous symmetry — is equivariant in the adjoint representation, which for SO(3) is $\mathfrak{so}(3)^* \cong \mathscr{V}_1$, so angular momentum lives in the ℓ=1 block. Since μ is **quadratic** and bilinear in the position–velocity pair, then — *provided the ℓ=1 block carries (at least) two copies of $\mathscr{V}_1$, e.g. $\mathscr{V}_1(pos) \oplus \mathscr{V}_1(vel)$, as in the model of Experiment 6* — the cross product is, up to scale, the **unique** SO(3)-equivariant degree-2 readout, and no degree-1 readout exists. *Proof.* Differentiate the equivariance of C_k and apply Schur (Hom_G between non-isomorphic isotypic types vanishes). For the bilinear μ on $\mathscr{Z} \supseteq \mathscr{V}_1(\mathrm{pos}) \oplus \mathscr{V}_1(\mathrm{vel})$, the degree-2 cross term in $\mathrm{Sym}^2\mathscr{Z}$ contains $\mathscr{V}_1 \otimes \mathscr{V}_1 = \mathrm{Sym}^2\mathscr{V}_1 \oplus \Lambda^2\mathscr{V}_1$; $\mathrm{Sym}^2\mathscr{V}_1 = \mathscr{V}_0 \oplus \mathscr{V}_2$ carries **no** $\mathscr{V}_1$, while $\Lambda^2\mathscr{V}_1 \cong \mathscr{V}_1$ with $\dim \mathrm{Hom}_{\mathrm{SO}(3)}(\Lambda^2\mathscr{V}_1, \mathscr{V}_1) = 1$ (classical SO(3) invariant theory) — realized by $r \times v$. So no degree-1 and no other degree-2 readout exists; uniqueness needs the two $\mathscr{V}_1$ copies (with m copies the space is $\binom{m}{2}$-dim). □

This *predicts*, with no fitted parameter, exactly what §5.3 measures — E from the invariant block ($R^2 \approx 1$), L from the ℓ=1 block via the degree-2 cross product (R^2=1.00) and nothing lower — and explains the tie to the companion paper's degree-1 Vector-Neuron cap (a degree-1 net *cannot* form the only available readout). **Scope (honest).** Proposition 4 is Schur's lemma applied to the symmetric powers plus the moment map's equivariance — a *placement principle*, not a new theorem: it pins which block carries each charge and at what degree, but representation theory alone does not force those charges to be the *dynamically slow* modes. What representation theory does not give, a one-line dynamical argument does — the *forward* direction of the hinge.

Proposition 5 (conservation ⇒ certified horizon — the slow side of the hinge). Let $Q : \mathscr{Z} \to W$ be a charge read-out and $\mathscr{K} \subseteq \mathscr{Z}$ a region forward-invariant under both the true dynamics Φ and the model f. Suppose Q is a first integral of the truth, $Q \circ \Phi = Q$ on $\mathscr{K}$, and the model conserves it up to a one-step defect $\eta := \sup_{z\in\mathscr{K}} \|Q(f(z)) - Q(z)\|_W$. Then the model's T-step error *in the charge value* is at most **linear** in the horizon,

$$\|Q(f^T z) - Q(\Phi^T z)\| = \|Q(f^T z) - Q(z)\| \;\leq\; T\,\eta \qquad (\forall z \in \mathscr{K},\ T \geq 0),$$

so the charge-readout channel carries **no positive Lyapunov exponent** ($\lambda_Q \leq 0$ in Theorem B's sense) and its certified horizon is $T_Q(\epsilon) \geq \epsilon/\eta$ — *infinite* at exact conservation $\eta = 0$ — irrespective of the ambient dynamics' largest Lyapunov exponent. *Proof.* Telescope $Q(f^T z) - Q(z) = \sum_{k<T}[Q(f(f^k z)) - Q(f^k z)]$; forward-invariance keeps each $f^k z \in \mathscr{K}$ so each term is $\leq \eta$, while Φ-invariance of $\mathscr{K}$ gives $Q(\Phi^T z) = Q(z)$. □

The content is the **additive-versus-multiplicative** contrast: a conserved charge's error *accumulates* ($T\eta$) where a generic channel's *compounds* ($e^{\lambda_j T}$, Theorem B). It is a statement about the **charge value**, not the full state — $f^T z$ and $\Phi^T z$ may separate at the ambient rate while their Q-images stay $T\eta$-close — which is exactly what a certificate for *"predict the conserved quantity"* needs (it does **not** claim the predictor-Jacobian's singular values on the conserved subspace are near one; off-diagonal shear can leave those large). With Proposition 4 this closes the **forward** direction of the hinge: Noether's charge μ lives in the invariant/equivariant blocks (Prop 4) **and** is certified to long horizons (Prop 5), so those blocks are the slow ones.

The hinge as a theorem, and its honest residue. Under the hinge's own hypothesis — that G is a symmetry of the *dynamics* — assemble the three pieces: if the latent flow is Hamiltonian with a G-invariant Hamiltonian, then (Noether) the moment map μ obeys $Q \circ \Phi = Q$; Proposition 4 places μ in $\mathscr{Z}_0 \oplus$ (adjoint); Proposition 5 certifies it. So *under symplectic structure* "the invariant/equivariant blocks are slow" is a **theorem**, not a coincidence. What stays assumed or measured, in order: **(i)** that the *learned latent* flow is Hamiltonian with G-invariant H — i.e. the encoder is essentially a symplectomorphism — is a structural assumption we do **not** enforce; **(ii)** exact conservation ($\eta = 0$) holds only for the momentum map under a G-equivariant *symplectic discretization* — energy is conserved only to $O(\Delta t^p)$ even by symplectic integrators (they preserve a *modified* Hamiltonian), and a generic learned f is neither symplectic nor equivariant, so η is **measured** (§5.3 is exactly that measurement); **(iii)** the **converse fails** — a slow channel need not be conserved (a rotation-invariant $|r|^2$ oscillates) — so the inclusion slow $\subseteq$ invariant $\oplus$ conserved-equivariant is one-directional. The upgrade over a pure conjecture is precise: *conserved* $\Rightarrow$ *slow* is now **proved** (Prop 5, at the charge-readout level), with the conservation defect η — not a positive Lyapunov exponent — bounding the drift *linearly*; only the dynamical-symmetry hypothesis and the value of η remain empirical.

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5. Experiments

All experiments are CPU/1-GPU-scale, run with explicit random seeds, and **gated**: a run reports `INCONCLUSIVE` rather than loosen a threshold. Load-bearing results are additionally guarded by unit tests. The experiment-to-code map, seed counts, and headline numbers are collected in Appendix A; multi-seed ranges below are seed min–max over three seeds unless noted.

5.1 The configuration axis

Experiment 1 (exact compositional flatness). On a structured object-interaction world with $\mathrm{SE}(3) \times S_n$ symmetry, the equivariant model's whole-pipeline relMSE is **exactly** $\times 1.00$ **flat over composition words** $m = 0, \ldots, 8$ (constant 0.2625), while a matched non-equivariant baseline blows up by $\times 12.89$. The predictor's Jacobian spectrum is measured directly: 48 **of** 96 **channels are contractive** ($\sigma \leq 1$, certified long-horizon) versus 48 expansive ($\max \sigma = 49$, $\min \sigma = 0.003$) — the clean coarse/fine split that is exactly Theorem B's input. A unit test verifies Theorem A holds *architecturally* at initialization (word-deviation 1.2×10^{-7}, before any training) while the control deviates 20%.

Experiment 2 (exponential certification on $\mathbb{Z}_2^6$). The 64 hexagrams of the *I Ching* are exactly $\{\pm 1\}^6 = \mathbb{Z}_2^6$, with the 6 changing-lines as a generating set. Training a sign-equivariant per-line predictor $f(z, a)_i = h_\theta(a_i) \odot z_i$ on **only the** 6 **single-line generators** (plus the identity, 7 of 64 actions) certifies it over **all** 2^6=64 **compositions** to machine precision (worst relMSE $\sim 10^{-33}$; equivariance residual exactly 0), while a non-equivariant baseline trained on the same 7 actions degrades monotonically with composition length (1.6–1.7×10^{-5} on seen actions $\rightarrow$ 0.59 at six flips, Figure 2). This makes "k generator checks $\Rightarrow 2^k$ certified set" literal, with genuine learning and an honest contrast.

5.2 The horizon × resolution staircase

Experiment 3 (the predictability staircase). A learned one-step predictor on a designed multi-Lyapunov system — a conserved invariant ($\lambda = -\infty$), a contracting channel ($\lambda = \ln 0.75$), a neutral SO(2) rotor ($\lambda = 0$), and a chaotic angle-doubling channel $z \mapsto z^2$ ($\lambda = \ln 2$) — is rolled out autoregressively. The model **recovers the chaotic Lyapunov exponent to within** 0.4% ($\hat{\lambda} = 0.690$–0.692 across three seeds, versus $\ln 2 = 0.693$), and its per-channel certified horizon obeys $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$: the chaotic "fine detail" is certifiable for only 3–10 steps over the full ϵ grid, and its horizon grows *only logarithmically* as ϵ coarsens (measured slope $\mathrm{d}T/\mathrm{d}\log\epsilon = 1.3$–$1.6 \approx 1/\lambda = 1.44$), while the **contracting channel stays certified for the entire** 90**-step rollout at every** ϵ (the conserved and rotor channels too, except at the finest ϵ=0.005, where they fall to $\sim$ 40–90 steps). At ϵ=0.05 the slow channels outlast the chaotic detail by 12–15× (Figure 3). This is the predictability staircase: long horizon $\Rightarrow$ coarse and invariant only.

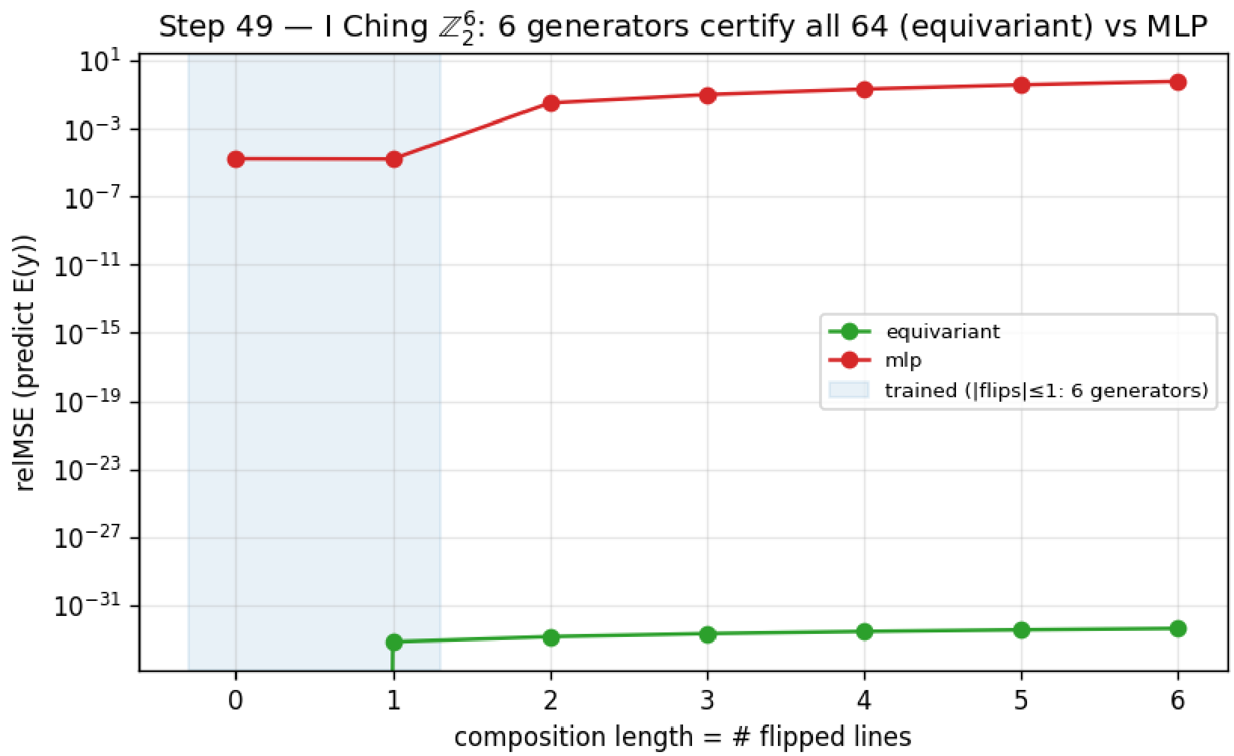

Figure 2: Training on the 6 generators of $\mathbb{Z}_2^6$ certifies all 64 compositions (equivariant, machine precision) while a non-equivariant baseline degrades with composition length.

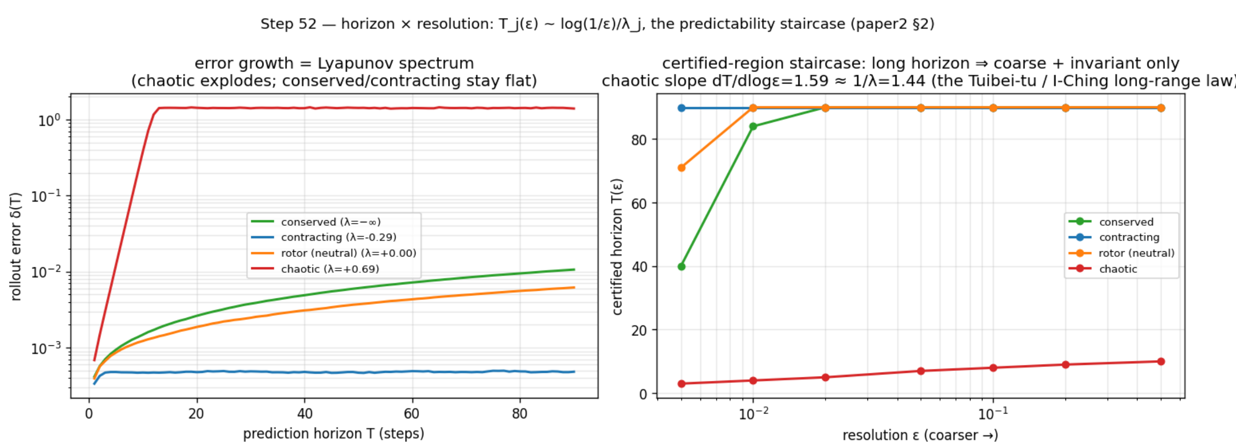

Figure 3: Left: rollout error growth recovers the Lyapunov spectrum. Right: the certified-horizon staircase $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$.

5.3 The Noether hinge

Experiment 4 (the hinge in 2D). On a two-dimensional SO(2) central-force system — conserved energy E and angular momentum L are invariant scalars, the orbital phase is fast — a learned equivariant autoencoder-with-latent-dynamics (a hand-built 2D Vector-Neuron model) spontaneously organizes the conserved quantities into its architecturally-invariant (ℓ=0) block. Regression R^2 for (E, L) is 0.92–0.99 **from the invariant block versus** ≤ 0.012 **from the equivariant (**ℓ**=1) block**, and the slowest mode the invariant block admits (0.009–0.031 across seeds) is 4.7–17× slower than anything the equivariant block admits (0.145, exactly the orbital phase velocity $2\sin(\omega\Delta t/2)$). We read the containment off this min-mode gap — the equivariant block's *slowest* achievable mode is already fast, so no slow direction escapes the invariant block in this system — rather than by a direct subspace test. The payoff is the certificate: the equivariant model's slow subspace **is** its invariant subspace, whose out-of-distribution group-action residual is $\sim 10^{-16}$, exact and architectural for all of SO(2). A matched non-equivariant baseline also learns E, L ($R^2 = 0.99$) but smears them across directions that **drift by** ≈ 1.17 under the same group action, and carries no certificate (Figure 4). The honest non-converse is visible in the data (the invariant but fast scalar $|r|^2$): slow $\subseteq$ invariant, not $=$.

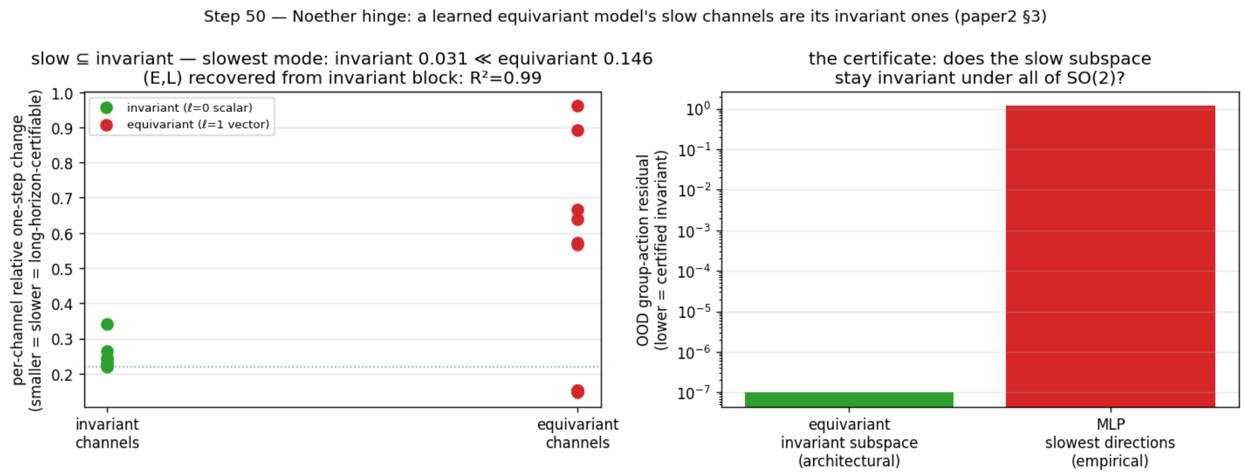

Figure 4: Left: the slowest latent modes live in the invariant (ℓ=0) block. Right: the certificate — the invariant subspace stays group-invariant to 10^{-16} where the non-equivariant model's slow directions drift by ~ 1.

Experiment 5 (lift to a 3D contact interaction). Pushing the hinge to a more embodied regime — two bodies in a three-dimensional well with a **soft pairwise repulsion (contact)**, modeled by an SO(3)-equivariant ℓ=0 $\oplus$ ℓ=1 Vector-Neuron — gives a *partial* lift, honestly reported. The *Noether-content* half lifts: the invariant ℓ=0 block recovers the conserved (E, contact-distance) at R^2=0.60–0.86 versus ≤ 0.06 for the ℓ=1 block, even with contact. But the clean containment slow $\subseteq$ invariant does **not** lift (invariant slowest mode 0.024 $\approx$ 0.026 for the equivariant block; this gate is reported `INCONCLUSIVE`) — for a principled reason: in 3D, angular momentum $L = \sum_i r_i \times v_i$ is a conserved ℓ=1 *vector*, so the equivariant block too carries a slow conserved mode. The clean 2D containment (where L is a scalar) becomes, in 3D, slow $\subseteq$ (invariant $\oplus$ conserved-equivariant). The Noether-content claim is dimension-robust; the clean-containment form is 2D-specific — a sharpening of scope, not a failure.

Experiment 6 (3D-aware containment). The conserved physics splits by isotypic *type* **and** polynomial *degree*: E (an invariant quadratic) is recovered linearly from the ℓ=0 block (R^2=0.62–0.91 across seeds), whereas $L = \sum_i r_i \times v_i$ (a conserved ℓ=1 vector) is **bilinear** — not linear in either block (≤ 0.04) but recovered at R^2=1.00 (range 0.998–1.000, the robust load-bearing result) by the **degree-two cross-product** readout of the ℓ=1 block. So slow $\subseteq$ (invariant $\oplus$ conserved-equivariant) holds exactly in 3D, with the equivariant conserved part accessed at degree two — and $r \times v$ is exactly the cross product a degree-one Vector-Neuron cannot form, tying the hinge's 3D form to the companion paper's bilinear-message result. The full gate, which additionally demands E-recovery R^2>0.8 in the ℓ=0 block, passes on one of three seeds — as honestly surfaced as the `INCONCLUSIVE` gate of Experiment 5 — while the degree-two L recovery is robust across all three.

5.4 Structure versus scale

Experiment 7 (structure versus scale). Train on a 50° wedge of an $\mathrm{SO}(2)$ orbit and test around the full circle. The exactly-equivariant model's error is **flat over the entire orbit** (out-of-wedge / in-wedge ratio 1.1–1.2) — a certificate that holds from partial data by the identity $\mathrm{err}(R_\theta s) = \mathrm{err}(s)$. A non-equivariant baseline **scaled across an** 88× **parameter range** (3.8k → 337k) buys genuine *interpolation*: its in-wedge error drops 31−166× and even **beats** the 56×-smaller equivariant model in-distribution — but out of the wedge the largest baseline remains 10–155× worse than the equivariant model and never reaches its floor (Figure 5). The fair nuance — *scale can win in-distribution* — is exactly why the contribution is the **certificate** (an orbit-wide guarantee from partial data), not a per-point accuracy contest. This is a gap versus *scale* (more parameters, the same data); the complementary question — whether *data augmentation* (orbit-covering rotations) closes it — is answered separately and honestly in §5.8 (it does, on a single orbit; the certificate's edge is then exactness, the a-priori guarantee, and worst-case optimality).

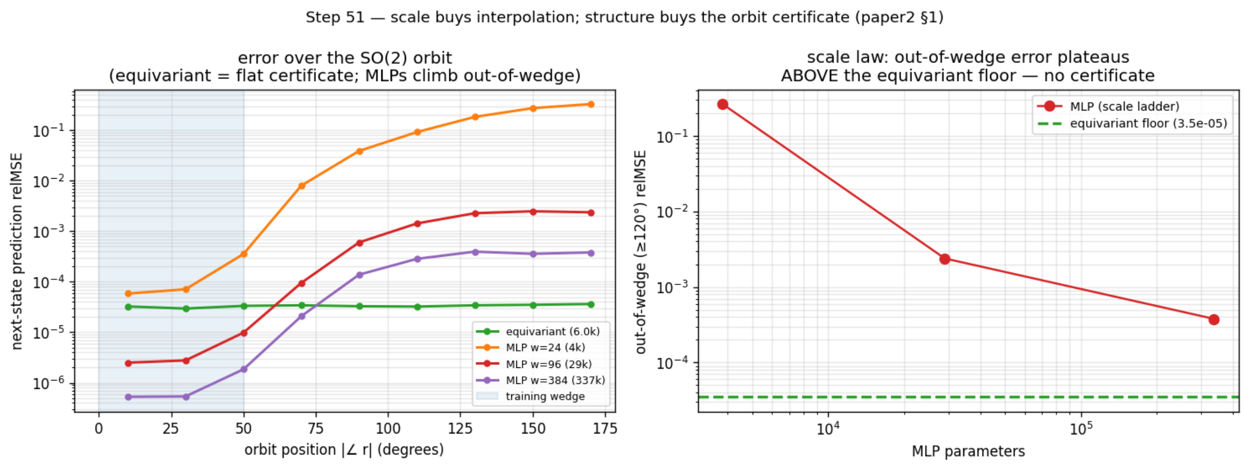

Figure 5: Left: error over the $\mathrm{SO}(2)$ orbit (equivariant flat; the scaled baselines dip below it in-wedge then climb out). Right: out-of-wedge error versus baseline scale plateaus far above the equivariant floor.

5.5 Approximate symmetry

Experiment 8 (graceful degradation and a measured threshold). We break the *world's* $\mathrm{SO}(2)$ symmetry with an anisotropy knob β (the potential becomes $V = \frac{1}{2}(x^2 + (1 + 2\beta)y^2)$, so angular momentum is no longer conserved) and re-run the wedge-train / full-circle-test. Three findings, robust across seeds: (i) at $\beta=0$ the certificate is exact — the equivariant model is 68–320× **better out-of-wedge** than the baseline; (ii) as the measured world symmetry-defect $\epsilon_{\mathrm{world}}$ grows, the equivariant out-of-wedge error grows **smoothly and monotonically** (correlation 0.88–0.98, exactly Theorem B's $\epsilon_{\max}$ term — a graceful slope, not a cliff); and (iii) the equivariant model keeps beating the non-equivariant one up to a **measured symmetry-content threshold** ($\epsilon_{\mathrm{world}} \approx 0.01$–$0.06$, seed-dependent), beyond which the now-wrong symmetry assumption hurts more than it helps (Figure 6). So *approximate* symmetry buys an *approximate* certificate with exactly the error budget the theory predicts, and the boundary where structure stops paying is itself measured, not assumed.

5.6 From certificate to action

A certificate is most useful if one need neither know the group in advance nor a decoder to act on it. Two results, re-framed from a companion line of experiments (not fresh runs here), close that loop. **Discovery:** from a blank slate of K learnable 3×3 matrices — nothing antisymmetric or bracket-closing imposed — gradient descent *rediscovers* a frozen teacher's symmetry algebra ($\mathfrak{so}(3)$, dimension 3, on a rotationally-symmetric teacher; the smaller $\mathfrak{so}(2)_z$, dimension 1, on a rotation-broken one — it reports the *surviving* group and refuses to invent the rest), and distilling those discovered generators into a free predictor buys back most of the across-group generalization the hard-wired equivariant model gets for nothing (closing more than half the out-of-distribution gap, matching the hand-wired oracle, transferring *exactly* the symmetry discovered). So the generating set S that anchors the configuration axis need not be postulated — it can

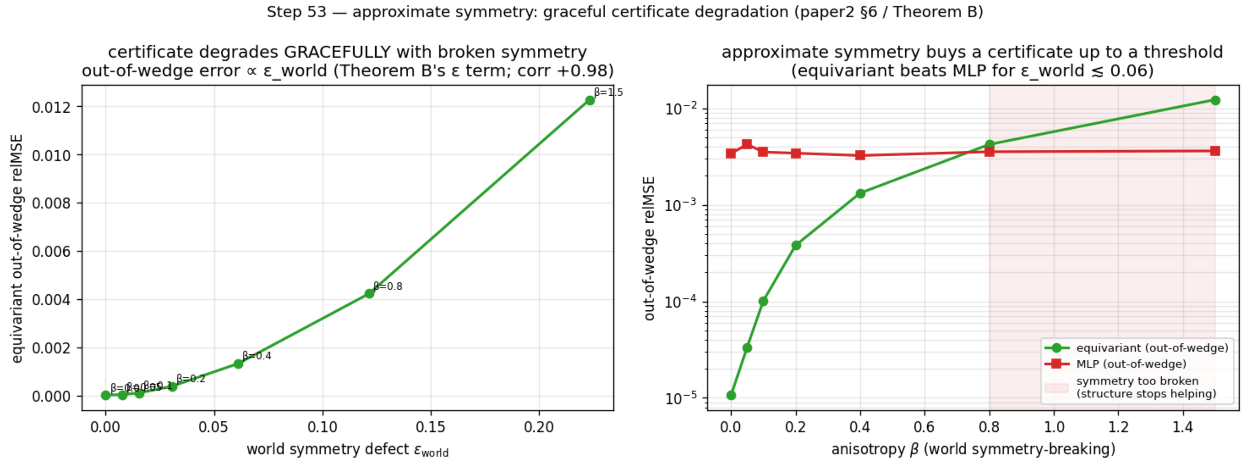

Figure 6: Left: equivariant out-of-wedge error grows $\propto \epsilon_{\text{world}}$ (Theorem B's ϵ term). Right: the equivariant model beats the baseline out-of-wedge up to a symmetry-content threshold, then crosses over.

be *measured*, falsifiably. **Generation:** given S, one can traverse the certified orbit to an unseen composed goal and execute it **closed-loop without a decoder** — latent-goal reaching driven purely by the equivariance identity over 24 paired seen-vs-unseen SE(3)-orbit tasks, closing ~ 0.59 of the orientation gap on the best deployable variant and — the load-bearing point — achieving the *same* success on the unseen orbit as on the seen one (unseen/seen ratio 1.000): *exact transfer*, not high absolute success. Together: *discover S → certify $\langle S \rangle$ → generate → act* — the certificate is not merely descriptive but a usable plan-and-execute criterion. These two results are re-framed from a companion line; §5.9 closes the same loop on a *fresh* run, with a planner driving a learned model of **real physics-engine contact dynamics**.

5.7 Beyond toys: the certificate on real contact dynamics

Experiment 9 (PushT, real physics-engine contact dynamics). Every experiment so far uses dynamics we *constructed* to be equivariant — the standing concern that the symmetry is built into a toy. We close that gap on **PushT**, a planar pushing benchmark whose contact dynamics are produced by a 2D physics engine we did not author. In the interior (the block kept away from the workspace walls) the dynamics are genuinely SO(2)-equivariant — rotating the whole scene rotates the physics. On a $\pm 50°$ wedge of real interior transitions we train an SO(2)-equivariant world model — an invariant-scalar-gated Vector-Neuron, exactly equivariant to residual ~ 10^{-7}, so its contact features (agent–block distance and contact angle) live in the invariant pathway — and, as a *fair* baseline, non-equivariant MLPs across a 160× parameter ladder (1.7k → 272k); both are trained one-step with identical data, optimiser, and cosine schedule. We then roll each model out $H=10$ steps and measure rollout relMSE across the full orbit of scene orientations (rotating a held-out real test set, as in Experiment 7).

Three findings, robust across 3 seeds (Figure 7): (i) the equivariant model's rollout error is **exactly flat over the orbit** (out-of-wedge / in-wedge ratio 1.00 at every horizon) — the certificate, now on a *learned* model of *real* contact physics; (ii) it is **competitive in-distribution** (in-wedge relMSE 0.13–0.15 vs the best MLP's 0.14–0.19), so the certificate is not bought by being a worse predictor — "flat" is also "good" here; and (iii) **no MLP scale reaches the equivariant floor out-of-wedge** — every scale across the ladder stays 2.1–3.9× above it at $H=10$, and the 272k-parameter MLP (16× the equivariant model's size) — the *closest* of the baselines — is still 2.1–2.6× clear of the floor. (These three findings hold on all three seeds; the experiment's *auxiliary* `climb` sub-check — the largest MLP's own in→out ratio exceeding 2× — is brittle, met on 1/3 seeds, so we report the load-bearing floor-penalty rather than that sub-check, and the committed gate prints `INCONCLUSIVE` on it.) The gap is smaller than the toy's because a single PushT step is easy to predict in-distribution; the certificate's value is precisely that it survives the *rollout*, out of distribution, where scale does not follow. This is the keystone non-toy result: the configuration certificate holds for a learned model of dynamics we did not design.

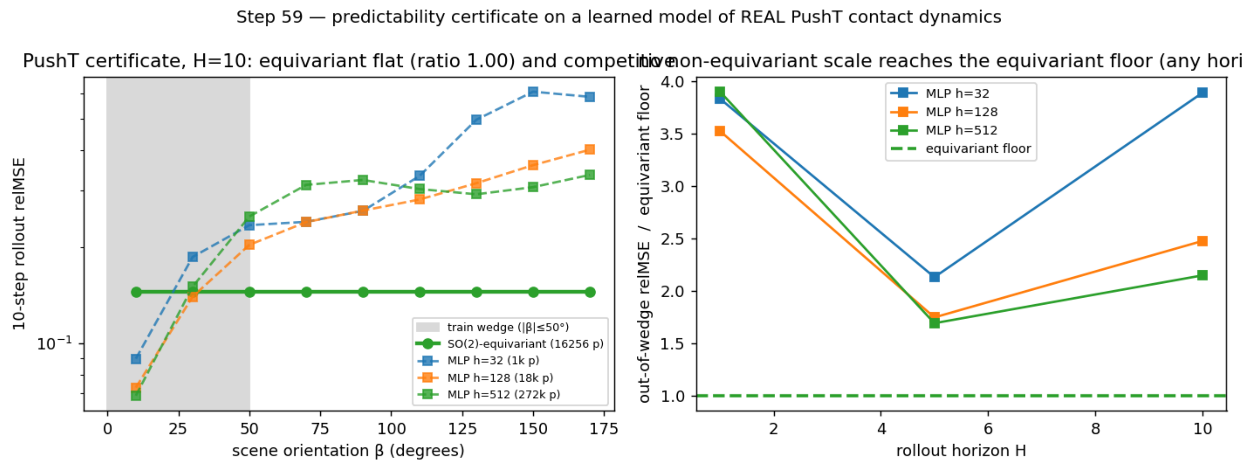

Figure 7: Experiment 9 (PushT, real contact dynamics). Left: 10-step rollout relMSE over the orbit of scene orientations — the learned SO(2)-equivariant model is exactly flat (ratio 1.00) and competitive in-distribution, while non-equivariant baselines dip in-wedge then climb out. Right: across a 160× parameter ladder no baseline reaches the equivariant floor out-of-wedge, at any rollout horizon.

5.8 Augmentation versus the certificate

Experiment 10 (the augmentation baseline — the sharpest objection, answered honestly). The strongest reply to §5.4/§5.7 is that a non-equivariant model with **data augmentation** (training on rotated copies) buys orbit-coverage cheaply, so the gap is an artifact of not augmenting. We test this on two axes and report what we find, not what we hoped.

Single orbit (PushT) — the concession. Adding an SO(2)-augmentation arm to the Experiment-9 protocol, the augmented MLP's 10-step rollout error becomes **flat over the orbit** (out/in ratio 0.93–1.02 across 3 seeds, versus the un-augmented MLP's 1.84–2.75). On a single continuous orbit, **augmentation does match the certificate** — we state this rather than hide it.

Composition axis ($\mathbb{Z}_2^6$). Here augmentation means training the non-equivariant MLP on more of the 64 words. The dynamics is smooth (bilinear $a \odot z$), so a well-trained MLP generalizes from ~half the words to a $\sim 10^{-4}$ floor — again, augmentation is a *strong* baseline. But it **never reaches the certificate's exactness**: the equivariant model is machine-exact ($\sim 10^{-32}$) from the 7 generators, $\sim 10^{28}\times$ below the MLP's approximation floor, and is *certified a priori* (Theorem A, Lemma 2) without ever testing the unseen compositions, whereas augmentation's success is only *confirmed by testing* the held-out words.

So the honest separation is **not** "structure generalizes, augmentation does not." It is that the certificate is **exact, a-priori, group-knowledge-free, and worst-case-optimal** (the ϵ/L-tube of §3.3 holds against *any* L-Lipschitz dynamics), whereas augmentation is an **approximate, post-hoc, group-informed average-case** improvement that ties on a benign single orbit and pays $O(\text{orbit})$ data for coverage the certificate gets from k generator checks. The gap of §5.4/§5.7 is specifically a gap versus *scale* (more parameters, same data); augmentation (more orbit-covering data) is the complementary axis, and the two together delimit exactly what structure buys that neither scale nor data does: a guarantee.

5.9 The certificate at the task level: closed-loop control

Experiment 11 (does the prediction certificate convert to task competence?). Experiment 9 certifies a model's *rollout error*; a sceptic rightly asks whether that proxy converts to *control*. We close the loop on the same real PushT physics, in a **contact-dominated pose task**: rotate the T-block to a target orientation (small translation only), so success depends on the block-rotation dynamics — exactly the channel where SO(2) bites, unlike a position-only push, which a near-linear agent subsystem carries out of distribution. We train the same invariant-scalar-gated equivariant model

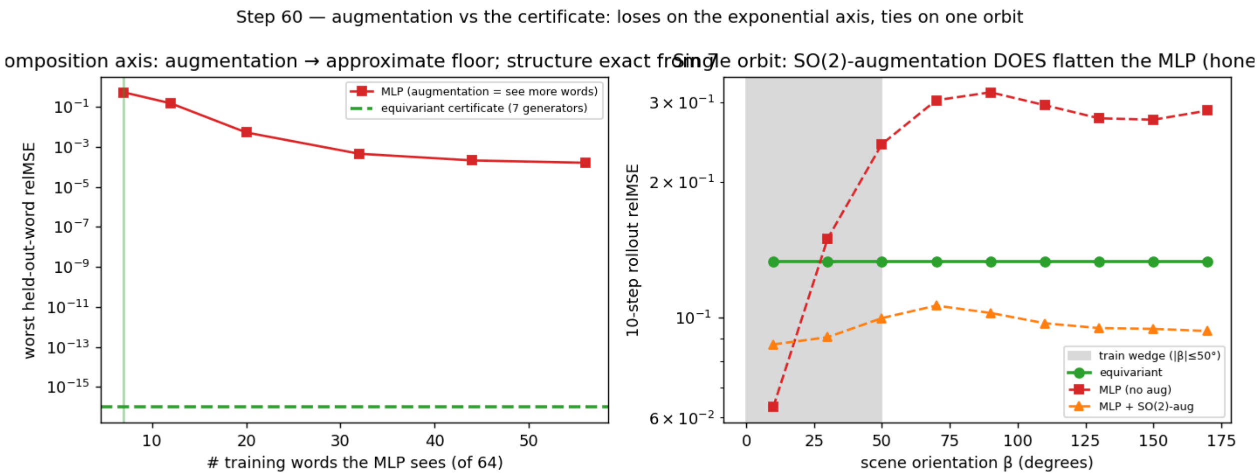

Figure 8: Experiment 10 (augmentation vs the certificate). Left: on $\mathbb{Z}_2^6$, augmentation (more training words) drives the MLP to a ~ 10^{-4} approximation floor, never the certificate's machine-exact ~ 10^{-32} from 7 generators. Right: on real PushT, SO(2)-augmentation flattens the MLP over the orbit (matching the certificate on a single orbit), unlike the un-augmented MLP.

and a scaled MLP baseline (4.3× its parameters) one-step on a wedge of real interior reorientation transitions, then run **CEM-MPC** on the real env across the full orbit of scene orientations, evaluating 24 base tasks *rotated to each orbit angle* — a **paired** protocol (the same task at every orientation) that removes the between-task variance which left earlier position-only closed-loop comparisons noise-limited.

The certificate's closed-loop clause (Theorem A under (A5)) needs a *G-equivariant planner*. We instantiate one: a CEM with an isotropic exploration covariance, a rotation-invariant **disk** action bound $\|a\| \leq 1$, and **scene-covariant** action noise. With these the planner satisfies $\pi(R_\beta s) = R_\beta\, \pi(s)$, so an equivariant model yields an equivariant closed-loop policy and the realized task error is *exactly* orbit-invariant.

Three findings (Figure 9): **(i) an exact certificate at the task level** — the model-rollout terminal pose error is flat over the orbit *to the float floor* (out-of-wedge / in-wedge ratio **1.000** on all 3 seeds — architectural) for the equivariant model, versus ×1.1–2.2 for the MLP under the *same* equivariant planner: the certificate now governs closed-loop *outcomes*, not just predictions; **(ii) it survives the real env** — executed on the real simulator the equivariant model's closed-loop block-angle error stays flat over the orbit to the measured precision (ratio **1.000** on all 3 seeds — the real interior physics is itself SO(2)-equivariant to ~ 10^{-5}/step, so the *realized* outcome, not just the prediction, is orbit-invariant) while the MLP degrades ×1.6–3.6 out of the wedge; and **(iii) flatness is not bought by being worse** — in-wedge the equivariant model is *competitive*, its block-angle error (3.6–16° across seeds) bracketing the MLP's (8–18°), each better on some seeds, so the orbit-flatness is that of a usable controller (we claim no in-distribution win).

The mechanism is the cost landscape itself: the pose cost the planner optimizes is SO(2)-invariant, and under the equivariant model its value is orbit-invariant to the float floor (drift $1.1\text{–}1.5\times10^{-7}$), while under the MLP it is materially distorted (drift 0.19–0.31, ~ 10^6× the equivariant floor). Two honesty notes, in the project's gating style. (a) That MLP cost-drift *straddles* a pre-registered absolute threshold (> 0.3, met on 1/3 seeds), so we report the eq/MLP *ratio* and the load-bearing orbit ratios — model-rollout ×1.000 and real-env ×1.000 versus the MLP's ×1.1–2.2 and ×1.6–3.6, on all 3 seeds — rather than the absolute drift, exactly as Experiment 9 reports its floor-penalty rather than the brittle `climb` sub-check. (b) Assumption (A5) genuinely *matters*: a scene-blind planner (raw, un-rotated CEM noise) keeps the equivariant model flatter on all 3 seeds (eq ×1.0–1.4 vs the MLP's ×1.5–3.5) but by a noisy margin — the *exact* guarantee needs the equivariant planner, while a realistic planner retains a softer version of the benefit. This is the certificate's strongest form — not a low error number but a **guarantee** that whatever competence holds in the wedge holds, zero-shot, over the entire orbit, to the float floor on every seed.

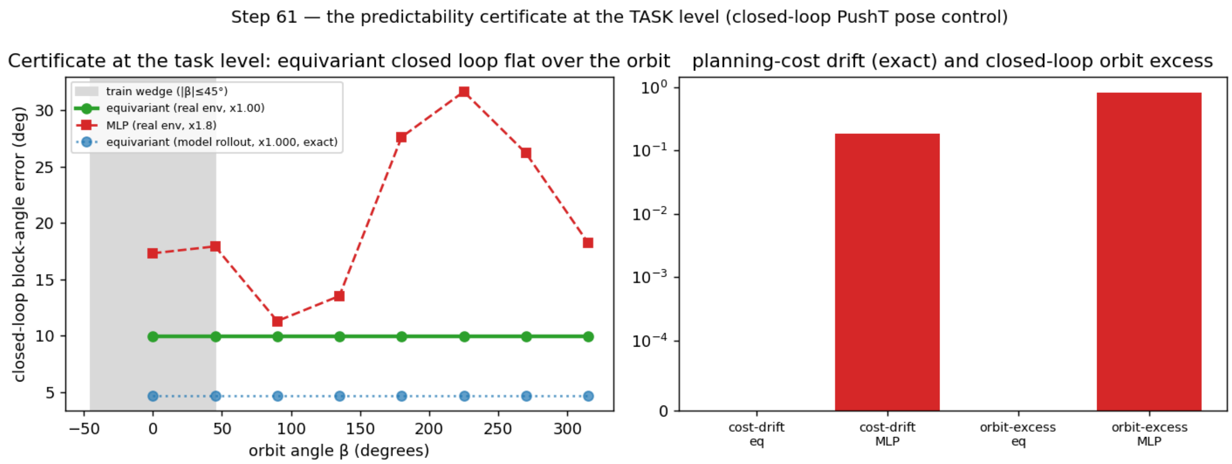

Figure 9: Experiment 11 (the certificate at the task level). Left: closed-loop block-angle error over the orbit of scene orientations — the equivariant model under a G-equivariant planner is exactly flat under model rollout (ratio 1.000) and flat on the real env, while a 4.3×-larger MLP degrades out of the training wedge. Right: the SO(2)-invariant planning cost is orbit-invariant to the float floor under the equivariant model and materially distorted under the MLP.

6. Related Work

The paper sits among a cluster of concurrent *structure-for-prediction* works. The table places each against the certificate's three axes and the *kind* of guarantee it offers (✓ provides; ~ partial/empirical; — not addressed); the prose elaborates below.

Work	Core mechanism	Config. $\langle S \rangle$	Horizon×ϵ	Closed-loop	Guarantee kind
BRo-JEPA	$\mathbb{Z}/10$ cyclic predictor	~ one group	—	—	empirical zero-shot
UWM-JEPA	$U(d)$ unitary predictor	~ one group	—	—	empirical
UR-JEPA / LeJEPA	latent-geometry prior (an/isotropy)	—	—	—	distributional (2nd-order)
IMWM	oracle-bypass; residual = search	—	—	~ search budget	diagnostic
LDA	SE(3)-intrinsic diffusion (action side)	—	—	~ policy	empirical, modest+robust
companion line	exact flatness: 1 element, 1 resolution	~ one element	—	✓ invariance	proved (a corner)
this paper	equivariant **predictability certificate**	✓ $k \Rightarrow \langle S \rangle$ (Thm A, Lem 1)	✓ $T_j(\epsilon) \sim \frac{\log(1/\epsilon)}{\lambda_j}$ (Thm B)	✓ exact (Exp 11)	**a-priori, per-situation, computable (Alg 1)**

The throughline: prior work supplies *mechanisms* (equivariant predictors), *priors* (latent geometry), or *diagnostics* (oracle-bypass) — each empirical or distributional and on a single axis; this paper supplies a *provable, computable region* across configuration × horizon × resolution, with the Noether hinge (§4) tying the axes together.

Equivariance and constant error (our companion line). A companion paper establishes, for a single group element and one resolution, that exact equivariance makes one-step error constant across the group, with exact-flatness and closed-loop-invariance corollaries. The present paper lifts that corner along three axes: Theorem A is its multi-step, full-monoid generalization; Theorem B adds the horizon × resolution stratification; and Lemma 1 turns k generator checks into an exponential certified set.

Equivariant predictors. Block-rotation predictors (BRo-JEPA, arXiv:2606.01372) match a $\mathbb{Z}/10\mathbb{Z}$ cyclic structure to the latent — "add k" becomes "rotate $k\theta$" — and report 99.46% best-config zero-shot transfer on MNIST modular arithmetic; unitary-predictor world models (UWM-JEPA, arXiv:2605.25313) do the analogous thing with $U(d)$. Theorem A explains *why* such zero-shot transfer occurs (the representation cancels inside the norm — orthogonal for the former, unitary for the latter), and the closure lemma quantifies *how far* it reaches (the whole generated monoid from its generators). These works share our toy-construction caveat, and we cite them as cross-group mechanism corroboration rather than scale results.

Latent-geometry regularizers. LeJEPA (Balestriero and LeCun, 2025) pushes the latent toward an isotropic Gaussian; UR-JEPA (Le, 2026; arXiv:2606.01443) challenges that target, arguing isotropy conflicts with the manifold hypothesis and that a data-discovered anisotropy is preferable. Our latent's anisotropy is neither isotropic nor data-discovered but **group-prescribed**: the covariance is pinned to the representation, $\rho\Sigma\rho^\top = \Sigma$ (measured residual 3×10^{-4}, versus 1.04 for a non-equivariant baseline). In a head-to-head (companion line, single seed) the data-fit anisotropy is best *in-distribution*, but only the group-prescribed anisotropy transfers out of distribution (a ~ 320× gap, consistent with the three-seed 68–320× and 10–155× of Experiments 8 and 7). The certificate is therefore complementary — a first-order, per-situation guarantee from the group, not a second-order distributional prior — and it predicts precisely *when* a discovered anisotropy will fail to generalize (off the training orbit).

Oracle-bypass diagnostics. IMWM (Gao et al., 2026; arXiv:2606.01626) uses the same falsifiable move as our capacity-ladder experiments — replace the learned model with oracle dynamics and see what is still missing — and finds a finite-budget planner *still* fails. It attributes the residual to **search** (proposal-sampling volume), whereas our analogous diagnosis attributes it to **representation** (permutation-invariant encoder pooling). These are complementary bottlenecks: a predictability certificate bounds the *model's* error over $\langle S \rangle \times T \times \epsilon$, and makes no claim about the planner's search budget downstream.

Geometry on the action side. LDA ("The Lie We Tell", Chuang et al., 2026; arXiv:2606.01847) names the *Euclidean Fallacy* — flattening an $\mathrm{SE}(3)$ pose into $\mathbb{R}^{12}$ breaks the manifold constraint, the coordinate-change equivariance, and geodesic optimality — and fixes it by running diffusion *on* $\mathrm{SE}(3)$ (tangent-space score, exponential-map retraction). It states our core motivation — *do not flatten geometric quantities* — on the action side. As a diffusion policy its gains are modest but robust (CALVIN average task length 3.27 → 3.51, +7.3%), the same profile as equivariant methods generally; our framing explains why a geometric prior buys a *kind* of guarantee (consistency across the group, robustness out of distribution) rather than a uniform accuracy jump.

Predictability horizons. The $T(\epsilon) \sim \log(1/\epsilon)/\lambda$ law is classical for dynamical systems (Lyapunov; numerical weather prediction). Our contribution is to measure it on a *learned latent world model*, tie its slow subspace to the group-invariant subspace via the Noether hinge, and fold it into a single multi-axis certificate for equivariant models.

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7. Limitations

- **Scale and scope.** All experiments are CPU/1-GPU. Experiments 9 and 11 validate the central claim on a real physics-engine contact simulator (PushT) — dynamics we did not author — at the *prediction* level (Experiment 9) and the closed-loop *task* level (Experiment 11, where the certificate becomes orbit-invariant pose control: out of the training wedge the equivariant controller stays flat while a scaled baseline degrades, the two being *comparable* in-distribution — we claim no in-distribution win). This is still *structured state* (not pixels) and a single $\mathrm{SO}(2)$ group; the out-of-distribution *prediction* gap is *modest* (2–4×, large only because a single PushT step is easy to predict in-distribution); and the closed-loop *out-of-wedge* advantage is shown on the contact-dominated pose task specifically — a position-only push stays a tie, since a near-linear agent subsystem carries it out of distribution. We claim a mechanism and a tool, demonstrated cleanly, not a scaled benchmark.

- **Scope of the exact certificate.** Theorem A requires (A3): the group must be a symmetry of the *dynamics*, not merely the encoder. The exact certificate therefore holds where the group is a genuine dynamical symmetry (orbital and conservative systems, free space, idealized manipulation). Everywhere else one is in Theorem B's approximate regime, whose degradation we *measure* rather than assume: Experiment 8 shows it is graceful ($\propto \epsilon_{\text{world}}$) up to a measured threshold $\epsilon_{\text{world}} \approx 0.01$–$0.06$ (seed-dependent; one seed crosses over as early as $\epsilon \approx 0.008$). The honest reading is "exact on symmetric dynamics; gracefully approximate, with a measured boundary, elsewhere."
- **Theorem B is a bound on the form, not a tightness proof.** The constants c_j are not estimated, and the *isotypic* refinement (channels chosen by ℓ-type, which ties the spectrum to equivariance and the hinge) is asserted, not separately measured — Experiment 3 measures the scalar law and Experiment 1 splits contractive from expansive but not by ℓ-type.
- **The Noether hinge's forward direction is proved; its hypotheses are measured.** The *placement* of each conserved charge by isotypic type (Proposition 4 — energy in ℓ=0, angular momentum in the ℓ=1 block via the unique degree-2 cross product) is forced by representation theory, and the *conserved* ⇒ *slow* direction is now a theorem (Proposition 5: a charge conserved to one-step defect η has prediction error $\le T\eta$, linear not exponential — certified to all horizons at η=0). What remains *assumed or measured*: that the learned latent flow is Hamiltonian with a G-invariant Hamiltonian (so the symmetry is a *dynamical* symmetry — the encoder essentially symplectic); the value of the defect η (exact only for the momentum map under a G-equivariant symplectic discretization, $O(\Delta t^p)$ for energy, measured for a generic learned f); and the **non-converse** (slow $\not\Rightarrow$ conserved). Proposition 5 is moreover a statement about the *charge value*, not full-state prediction on the conserved subspace. Its lift to a *real, contact-rich embodied* model — approximate symmetry, latent learned end-to-end — is the primary open problem; the 3D contact experiment is a clean two-body toy whose clean-containment gate is `INCONCLUSIVE` (resolved in type-and-degree form by Experiment 6).
- **§5.6 re-frames companion-line results**, not fresh runs: the decoder-free reach is *exact transfer* (unseen/seen ratio 1.000) at a *partial* absolute success (~ 0.59 of the orientation gap). We report the transfer ratio as the load-bearing claim, not absolute task success.
- **"Flat" is not "good."** A constant error across the group says nothing about its magnitude; Theorem A certifies *consistency*, and the "scale never reaches the certificate" claim is an *architectural* fact, **proved** in Lemma 2 (orbit-constant error against every equivariant target $\iff$ equivariance) and quantified in §3.3 (a non-equivariant L-Lipschitz learner certifies only an ϵ/L-tube around the data) — not the conclusion of a finite sweep.

Falsifiability. The framing makes refutable predictions, not after-the-fact rationalizations. A finite non-equivariant network attaining exact orbit-flatness would break Theorem A's architectural premise; a symmetric-dynamics system whose conserved/slow modes lay *outside* the invariant ⊕ conserved-equivariant subspace would break the hinge; and a channel whose certified horizon failed to scale as $\log(1/\epsilon)/\lambda_j$ would break Theorem B. We have not observed these; they are the experiments that would sink the thesis.

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8. Conclusion

For equivariant world models, structure delivers a *kind* of result scaling cannot: a **predictability certificate** — a provable, computable, training-free region across configuration, horizon, and resolution. We established the master theorem (Theorem A, the exact configuration certificate, proved — its closed-loop clause under the assumed equivariant-planner condition; Lemma 2, its converse, characterizing the certificate *as* equivariance; Theorem B, the spectral horizon × resolution law, stated as a bound and measured), exhibited the certified region as the coarse-invariant-slow-low-composition corner, proved the Noether hinge's forward direction that unifies the symmetry and time axes (Proposition 5: conserved ⇒ slow) and measured its defect, and showed — empirically and via the quantitative separation of §3.3 — that an 88×-scaled non-equivariant model buys interpolation but never the certificate, an architectural impossibility rather than an unfinished sweep. *Scale buys interpolation; structure buys a certificate.* The same criterion that tells you which compositions a robot policy will handle zero-shot also tells you why eclipses are forecastable for millennia while weather is not — one structural law, read across domains.

References

Concurrent and prior work referenced above (arXiv identifiers from the project's source notes; external numbers are quoted from the cited works):

- **BRo-JEPA** — block-rotation $\mathbb{Z}/10\mathbb{Z}$ predictor; 99.46% best-config zero-shot on MNIST modular arithmetic. arXiv:2606.01372.
- **UWM-JEPA** — unitary-predictor ($U(d)$) belief-space world model. arXiv:2605.25313.
- **UR-JEPA** (Le, 2026) — uniform-rectifiability latent regularizer; manifold-hypothesis critique of isotropic SIGReg. arXiv:2606.01443.
- **IMWM** (Gao et al., 2026) — oracle-bypass diagnosis localizing the residual to planner *search*. arXiv:2606.01626.
- **LDA / "The Lie We Tell"** (Chuang et al., 2026) — the *Euclidean Fallacy*; SE(3)-intrinsic diffusion policy; CALVIN average task length 3.27 → 3.51 (+7.3%). arXiv:2606.01847.
- **LeJEPA** (Balestriero and LeCun, 2025) — SIGReg / isotropic-Gaussian self-supervised objective (the latent-geometry target that UR-JEPA, and our group-prescribed anisotropy, depart from).

The $T(\epsilon) \sim \log(1/\epsilon)/\lambda$ predictability-horizon law is classical (Lyapunov / Lorenz; standard numerical-weather-prediction practice).

Appendix A. Reproducibility

Every experiment sets random seeds explicitly, prints an `INCONCLUSIVE` verdict rather than loosen a gate, and writes its figure and JSON to `papers/figures/`. The multi-seed steps commit per-seed JSONs (`papers/figures/step5*_seeds.json`, `step59_pusht_certificate_seeds.json`, `step60_augmentation_seeds.json`, and `step61_closed_loop_certificate_seeds.json`, seeds 0/1/2), regenerated by `experiments/aggregate_seeds.py`; every range quoted above is the seed min–max from those files. The configuration-flatness experiment (Experiment 1) self-aggregates its three seeds into the means in `step47_certificate.json`. The full test suite passes together; `tests/conftest.py` isolates the float64 experiments from the float32 codebase. One command reproduces the paper end-to-end — `make paper2` (multi-seed re-runs → figures → tests → PDF), with `make paper2-quick` for the figures-and-PDF fast path; everything is CPU/MPS, no CUDA.

Exp.	Result	Code	Test	Seeds	Headline number
1	Exact compositional flatness + spectrum	`experiments/step47_certificate.py`	`tests/test_step47_certificate.py`	3	relMSE ×1.00 flat (m=0..8); 48/96 contractive
2	$\mathbb{Z}_2^6$ exponential certificate	`experiments/step49_iching_certificate.py`	—	1	6 generators → all 64; worst relMSE ~ 10^{-33}
3	Horizon × resolution staircase	`experiments/step52_horizon_resolution.py`	`tests/test_step52_horizon_resolution.py`	3	$\hat{\lambda}$=0.69 vs ln 2; slope ≈ $1/\lambda$
4	Noether hinge (2D)	`experiments/step50_noether_hinge.py`	`tests/test_step50_noether_hinge.py`	3	R^2 0.92–0.99 vs ≤ 0.012; cert 10^{-16} vs 1.17

Exp.	Result	Code	Test	Seeds	Headline number
5	Lift to 3D contact	`experiments/step57_embodied_hinge.py`	—	3	clean-containment gate **INCONCLUSIVE**; Noether content lifts (invariant R^2=0.60–0.86 vs $\leq$ 0.06); resolved by Exp. 6
6	3D-aware containment	`experiments/step58_3d_containment.py`	—	3	$E \to \ell$=0 linear (R^2=0.62–0.91); L bilinear $\to \ell$=1 degree-2 cross (R^2=1.00, range 0.998–1.000)
7	Structure vs scale	`experiments/step51_structure_vs_scale.py`	—	3	equiv flat 1.1–1.2; in-wedge gain 31–166×; best baseline 10–155× above floor out-of-wedge
8	Approximate symmetry	`experiments/step53_approximate_symmetry.py`	—	3	cert exact at β=0 (68–320×); graceful $\propto \epsilon$ (corr 0.88–0.98); threshold $\epsilon \approx$ 0.01–0.06
9	**Certificate on real contact dynamics (PushT)**	`experiments/step59_pusht_certificate.py`	—	3	learned equivariant exactly flat over the orbit (ratio 1.00, equiv-resid $\sim 10^{-7}$) and competitive in-dist; no MLP scale (1.7k–272k) reaches the floor out-of-wedge (2.1–3.9×, H=10)

Exp.	Result	Code	Test	Seeds	Headline number
10	**Augmentation vs the certificate**	`experiments/step60_augmentation.py`		3	SO(2)-aug flattens a single PushT orbit (ratio 0.93–1.02 vs plain 1.84–2.75); on $\mathbb{Z}_2^6$ augmentation reaches a $\sim 10^{-4}$ floor but never the certificate's exact $\sim 10^{-32}$ from 7 generators ($\sim 10^{28}\times$)
11	**Certificate at the task level (closed-loop PushT pose control)**	`experiments/step61_closed_loop_certificate.py`	`tests/test_step61.py`	3	equivariant model + G-equivariant planner: closed-loop pose error orbit-invariant to the float floor (model-rollout & real-env ratio 1.000, all seeds) vs MLP ×1.1–2.2 (rollout) / ×1.6–3.6 (real-env); in-wedge competitive (3.6–16° vs 8–18°). Auxiliary cost-drift > 0.3 met 1/3 (eq $\sim 10^{-7}$ vs MLP 0.19–0.31)